Embedded Systems Internship Project

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Internship Guide: Mrs. Neha Jamdar

AICTE - Microchip Virtual Internship

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Project Overview

Project Title

Design and Simulation of Battery Management System using PIC16F877A Microcontroller

Core Objective

Develop a low-cost microcontroller-based system for monitoring and protecting batteries from overvoltage and undervoltage conditions

Implementation

Simulation-based approach using Proteus software with potentiometer-based battery emulation

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Why Battery Management Systems Matter

Critical Applications

- Electric vehicles
- Renewable energy storage
- Portable electronics
- Industrial systems

Key Benefits

BMS ensures safe battery operation by continuously monitoring voltage, temperature, and charge levels. It prevents damage from overcharging and deep discharge, extending battery life and preventing safety hazards.

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Problem Statement

Overcharging Risk

Excessive voltage can damage battery cells and reduce lifespan

Deep Discharge

Undervoltage conditions can permanently damage battery chemistry

Safety Hazards

Improper usage can lead to overheating, system failure, or explosion

The challenge: Design a **low-cost, microcontroller-based system** that monitors battery voltage, detects abnormal conditions, and provides appropriate control signals.

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Project Objectives

01

Design BMS Architecture

Implement system using PIC16F877A microcontroller

02

ADC Voltage Monitoring

Read analog battery voltage through ADC module

03

Threshold Detection

Compare values with overvoltage and undervoltage thresholds

04

Generate Control Signals

Activate LEDs and buzzer for different conditions

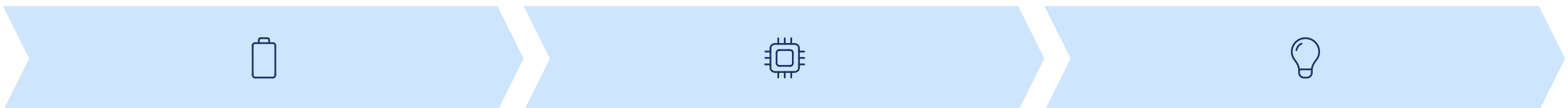
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Simulate Battery Behavior

Use potentiometer in Proteus to emulate varying voltage levels

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System Architecture



Input

Potentiometer simulates battery voltage connected to ADC pin

Processing

PIC16F877A reads ADC, compares thresholds, executes logic

Output

LEDs and buzzer indicate normal, overvoltage, or undervoltage states

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Working Logic

Overvoltage Condition

Voltage exceeds threshold → RED LED and Blue LED 1 (Allow Charging) activate

Undervoltage Condition

Voltage below threshold → BUZZER sounds and Blue LED 2 (Allow Discharging) activates

Normal Operation

Within safe range → GREEN LED and both Blue LEDs illuminate

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Tools and Technologies



MPLAB X IDE

Code development environment for microcontroller programming



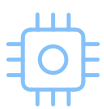
XC8 Compiler

Compilation tool for generating HEX files



Proteus Design Suite

Circuit simulation and virtual testing platform



PIC16F877A

8-bit microcontroller with ADC module for voltage sensing

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Simulation Results

Undervoltage Fault

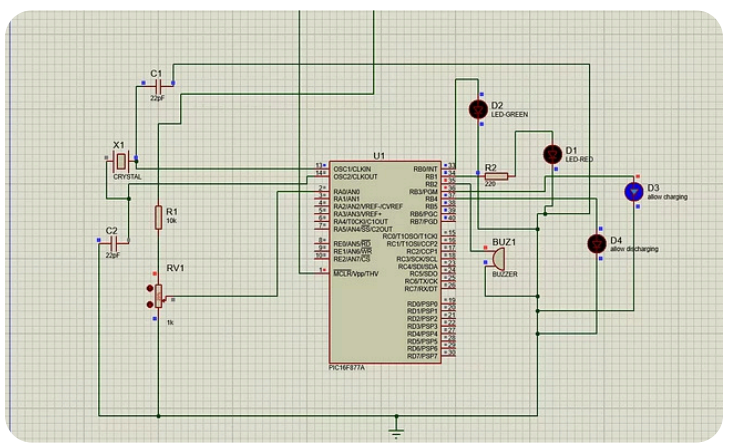
Buzzer activated with Blue LED 2, indicating charging allowed but discharge is prohibited

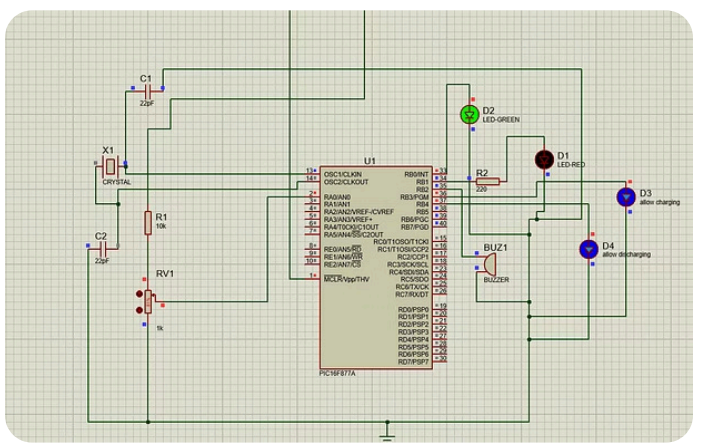
Normal Operation

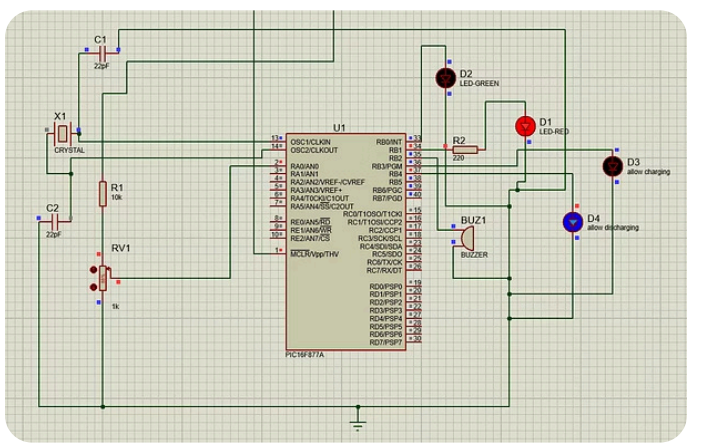
GREEN LED illuminated with both Blue LEDs, indicating safe voltage range

Overvoltage Fault

RED LED activated with Blue LED 1, indicating discharging allowed but charging prohibited







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Key Achievements and Future Scope

Technical Skills

- Embedded systems programming
- Microcontroller ADC interfacing
- Proteus simulation expertise

Project Success

- Functional BMS simulation
- Real-time voltage monitoring
- Control signal implementation

Future Applications

- Electric vehicle battery systems
- Solar energy storage
- IoT-based monitoring

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